In [ ]:

```
8students = []


def calculate_result(student):
    total = (
        student["python"]
        + student["artifical intelligence"]
        + student["mathematics"]
    )

    percentage = (total / 300) * 100

    if percentage >= 90:
        grade = "A+"
    elif percentage >= 80:
        grade = "A"
    elif percentage >= 70:
        grade = "B"
    elif percentage >= 60:
        grade = "C"
    else:
        grade = "F"

    student["total"] = total
    student["percentage"] = percentage
    student["grade"] = grade


# Input for 5 students
for i in range(5):
    print("\nEnter the details of student", i + 1)

    name = input("Enter the name: ")
    roll_no = input("Enter the roll no: ")
    python = int(input("Enter the marks in Python: "))
    artifical_intelligence = int(
        input("Enter the marks in Artificial Intelligence: ")
    )
    mathematics = int(input("Enter the marks in Mathematics: "))

    student = {
        "name": name,
        "roll_no": roll_no,
        "python": python,
        "artifical intelligence": artifical_intelligence,
        "mathematics": mathematics
    }

    calculate_result(student)
    students.append(student)


# Student with the highest marks
```

```python
highest_student = max(students, key=lambda student: student["total"])


# Display results
for student in students:
    print("\nName:", student["name"])
    print("Roll No:", student["roll_no"])
    print("Python:", student["python"])
    print("Artificial Intelligence:", student["artifical intelligence"])
    print("Mathematics:", student["mathematics"])
    print("Total:", student["total"])
    print("Percentage:", student["percentage"])
    print("Grade:", student["grade"])


print("\nStudent with the highest marks")
print("Name:", highest_student["name"])
print("Total:", highest_student["total"])


# Save results to file
with open("results.txt", "w") as file:
    file.write("STUDENT RESULTS\n")
    file.write("====================\n")

    for student in students:
        file.write(f"\nName: {student['name']}\n")
        file.write(f"Roll No: {student['roll_no']}\n")
        file.write(f"Python: {student['python']}\n")
        file.write(
            f"Artificial Intelligence: "
            f"{student['artifical intelligence']}\n"
        )
        file.write(f"Mathematics: {student['mathematics']}\n")
        file.write(f"Total: {student['total']}\n")
        file.write(f"Percentage: {student['percentage']:.2f}%\n")
        file.write(f"Grade: {student['grade']}\n")
        file.write("--------------------\n")

    file.write("\nHIGHEST MARKS\n")
    file.write("====================\n")
    file.write(f"Name: {highest_student['name']}\n")
    file.write(f"Total: {highest_student['total']}\n")

print("\nResults saved in results.txt")
3
```

```
Enter the details of student 1
Enter the name: asba
Enter the roll no: 23
Enter the marks in Python: 34
Enter the marks in Artificial Intelligence: 50
Enter the marks in Mathematics: 70

Enter the details of student 2
Enter the name: www
Enter the roll no: 24
Enter the marks in Python: 56
Enter the marks in Artificial Intelligence: 78
Enter the marks in Mathematics: 90

Enter the details of student 3
Enter the name: 34
Enter the roll no: 45
Enter the marks in Python: 300
Enter the marks in Artificial Intelligence: 9
Enter the marks in Mathematics: 90

Enter the details of student 4
Enter the name: wowow
Enter the roll no: 34
Enter the marks in Python: 800
Enter the marks in Artificial Intelligence: 10
Enter the marks in Mathematics: 200

Enter the details of student 5
Enter the name: rall
Enter the roll no: 19
Enter the marks in Python: 40
Enter the marks in Artificial Intelligence: 39
Enter the marks in Mathematics: 78

Name: asba
Roll No: 23
Python: 34
Artificial Intelligence: 50
Mathematics: 70
Total: 154
Percentage: 51.33333333333333
Grade: F

Name: www
Roll No: 24
Python: 56
Artificial Intelligence: 78
Mathematics: 90
Total: 224
Percentage: 74.66666666666667
Grade: B

Name: 34
Roll No: 45
```

```
Python: 300
Artificial Intelligence: 9
Mathematics: 90
Total: 399
Percentage: 133.0
Grade: A+

Name: wowow
Roll No: 34
Python: 800
Artificial Intelligence: 10
Mathematics: 200
Total: 1010
Percentage: 336.6666666666667
Grade: A+

Name: rall
Roll No: 19
Python: 40
Artificial Intelligence: 39
Mathematics: 78
Total: 157
Percentage: 52.33333333333333
Grade: F

Student with the highest marks
Name: wowow
Total: 1010

Results saved in results.txt
```

Out[ ]: 3

In [ ]:
```python
products = {}
def calculate_bill(products):
    total = 0

    for product in products:
        price = products[product]["Price"]
        quantity = products[product]["Quantity"]

        product_total = price * quantity
        total += product_total

    return total
# input
for i in range(5):
    print("\nEnter Product", i + 1)

    name = input("Product name: ")
    price = float(input("Price: "))
    quantity = int(input("Quantity: "))

    products[name] = {
        "Price": price,
```

```
        "Quantity": quantity
    }

# Display
print("\nSHOPPING BILL")

for product in products:
    price = products[product]["Price"]
    quantity = products[product]["Quantity"]

    product_total = price * quantity
    print(product, "=", price, "x", quantity, "=", product_total)


# Calculate complete bill
total_bill = calculate_bill(products)

print("\nComplete Bill:", total_bill, "Rs.")


# Apply discount
discount = 0

if total_bill > 10000:
    discount = total_bill * 0.10

final_bill = total_bill - discount

print("Discount:", discount, "Rs.")
print("Final Bill:", final_bill, "Rs.")


# expensive product
most_expensive = max(products,key=lambda x: products[x]["Price"])

print("\nMost expensive product:", most_expensive)
print("Price:", products[most_expensive]["Price"], "Rs.")

# Save
with open("bill.txt", "w") as file:
    file.write("SHOPPING BILL\n")

    for product in products:
        price = products[product]["Price"]
        quantity = products[product]["Quantity"]
        product_total = price * quantity

        file.write(product + " - " +str(price) + " x " +str(quantity) + "

    file.write("\nComplete Bill: " + str(total_bill) + " Rs.\n")
    file.write("Discount: " + str(discount) + " Rs.\n")
    file.write("Final Bill: " + str(final_bill) + " Rs.\n")
    file.write("Most Expensive Product: " + most_expensive + "\n")
```

```
print("\nBill saved in bill.txt")
```

```
Enter Product 1
Product name: bag
Price: 4000
Quantity: 2

Enter Product 2
Product name: trench coat
Price: 7000
Quantity: 5

Enter Product 3
Product name: ham burger
Price: 2000
Quantity: 1

Enter Product 4
Product name: shampoo
Price: 3000
Quantity: 3

Enter Product 5
Product name: tea
Price: 1000
Quantity: 2

SHOPPING BILL
bag = 4000.0 x 2 = 8000.0
trench coat = 7000.0 x 5 = 35000.0
ham burger = 2000.0 x 1 = 2000.0
shampoo = 3000.0 x 3 = 9000.0
tea = 1000.0 x 2 = 2000.0

Complete Bill: 56000.0 Rs.
Discount: 5600.0 Rs.
Final Bill: 50400.0 Rs.

Most expensive product: trench coat
Price: 7000.0 Rs.

Bill saved in bill.txt
```

In [ ]:

```
def count_vowels(sentence):
    vowels = "aeiouAEIOU"
    count = 0

    for character in sentence:
        if character in vowels:
            count += 1

    return count

sentence = input("Enter a sentence: ")
words = sentence.split()
```

```
total_words = len(words)
total_characters = len(sentence)

longest_word = max(words, key=len)
shortest_word = min(words, key=len)
vowel_count = count_vowels(sentence)
unique_words = set(words)
alphabetical_words = sorted(unique_words)

# Display
print("\nSENTENCE ANALYSIS")

print("Total number of words:", total_words)
print("Total number of characters:", total_characters)
print("Longest word:", longest_word)
print("Shortest word:", shortest_word)
print("Number of vowels:", vowel_count)
print("Unique words:", unique_words)
print("Words in alphabetical order:")
for word in alphabetical_words:
    print(word)
# Save analysis to file
with open("word_analysis.txt", "w") as file:
    file.write("SENTENCE ANALYSIS\n")
    file.write("Sentence: " + sentence + "\n")
    file.write("Total number of words: " + str(total_words) + "\n")
    file.write("Total number of characters: " + str(total_characters) + "
    file.write("Longest word: " + longest_word + "\n")
    file.write("Shortest word: " + shortest_word + "\n")
    file.write("Number of vowels: " + str(vowel_count) + "\n")
    file.write("Unique words: " + str(unique_words) + "\n")
    file.write("Alphabetical order: " + str(alphabetical_words) + "\n")

print("\nAnalysis saved in word_analysis.txt")
```

```
Enter a sentence: my name is asba

SENTENCE ANALYSIS
Total number of words: 4
Total number of characters: 16
Longest word: name
Shortest word: my
Number of vowels: 5
Unique words: {'is', 'name', 'asba', 'my'}
Words in alphabetical order:
asba
is
my
name

Analysis saved in word_analysis.txt
```

In [ ]:
```
menu = {
    1: {"name": "Burger", "price": 500},
    2: {"name": "Pizza", "price": 800},
```

```
    3: {"name": "Fries", "price": 250},
    4: {"name": "Drink", "price": 150}}

def calculate_bill(order):
    total = 0

    for item in order:
        price = menu[item]["price"]
        quantity = order[item]

        total += price * quantity

    return total
order = {}
while True:

    print("\nRESTAURANT MENU")
    print("1. Burger   Rs. 500")
    print("2. Pizza    Rs. 800")
    print("3. Fries    Rs. 250")
    print("4. Drink    Rs. 150")
    print("5. Finish Order")

    choice = int(input("Select an item: "))

    if choice == 5:
        break

    if choice in menu:
        quantity = int(input("Enter quantity: "))

        # Add quantity
        if choice in order:
            order[choice] += quantity
        else:
            order[choice] = quantity

        print(menu[choice]["name"],"added to your order.")

    else:
        print("Invalid choice. Please select again.")


# Calculate
total_bill = calculate_bill(order)

# Display bill
print("\nFINAL BILL")

for item in order:
    name = menu[item]["name"]
    price = menu[item]["price"]
    quantity = order[item]
```

```
    item_total = price * quantity

    print(name,"- Rs.", price,"x", quantity,"=", item_total, "Rs.")
print("Final Bill:", total_bill, "Rs.")

# Save
with open("orders.txt", "w") as file:

    file.write("RESTAURANT BILL\n")

    for item in order:
        name = menu[item]["name"]
        price = menu[item]["price"]
        quantity = order[item]

        item_total = price * quantity

        file.write(name + " - Rs. " +str(price) + " x " +str(quantity) +

    file.write("--------------------------------\n")
    file.write("Final Bill: " + str(total_bill) + " Rs.\n")

print("\nBill saved in orders.txt")
```

```
RESTAURANT MENU
1. Burger   Rs. 500
2. Pizza    Rs. 800
3. Fries    Rs. 250
4. Drink    Rs. 150
5. Finish Order
Select an item: 3
Enter quantity: 2
Fries added to your order.

RESTAURANT MENU
1. Burger   Rs. 500
2. Pizza    Rs. 800
3. Fries    Rs. 250
4. Drink    Rs. 150
5. Finish Order
Select an item: 5

FINAL BILL
Fries - Rs. 250 x 2 = 500 Rs.
Final Bill: 500 Rs.

Bill saved in orders.txt
```

In [ ]:

```
def addition(a, b):
    return a + b
def subtraction(a, b):
    return a - b
def multiplication(a, b):
    return a * b
def division(a, b):
```

```
    if b == 0:
        return "Cannot divide by zero"
    return a / b
def modulus(a, b):
    if b == 0:
        return "Cannot find modulus with zero"
    return a % b
while True:

    print("\nCALCULATOR")

    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))

    print("\nSelect an operation:")
    print("+ Addition")
    print("- Subtraction")
    print("* Multiplication")
    print("/ Division")
    print("% Modulus")

    operation = input("Enter operation: ")

    if operation == "+":
        result = addition(num1, num2)
    elif operation == "-":
        result = subtraction(num1, num2)
    elif operation == "*":
        result = multiplication(num1, num2)
    elif operation == "/":
        result = division(num1, num2)
    elif operation == "%":
        result = modulus(num1, num2)
    else:
        result = "Invalid operation"
    print("Result:", result)
    again = input("\nDo you want to perform another calculation? (yes/no)

    if again.lower() != "yes":
        print("Calculator closed.")
        break
```

```
CALCULATOR
Enter first number: 61
Enter second number: 34

Select an operation:
+ Addition
- Subtraction
* Multiplication
/ Division
% Modulus
Enter operation: *
Result: 2074.0

Do you want to perform another calculation? (yes/no): no
Calculator closed.
```

In [ ]: